Son Nguyen

ngtrtson@unc.edu | github.com/nttson | linkedin.com/in/snguyen3 | nttson.github.io

Education

University of North Carolina–Chapel Hill – BS in Computer Science — Expected 2029

Arizona State University – BS in Computer Science — 2025 – 2026

Barrett, The Honors College student; transferred to UNC-CH — 4.16 GPA

Tran Dai Nghia High School for the Gifted – Ho Chi Minh City, Vietnam — Graduated May 2025

Computer Science Specialization Program; Class Entrance Valedictorian; Class President

Publications

[3] **Son Nguyen**, V. G. Bao, Quang Minh Phan, Trong P. Le. The Shape of Noise: Layer-Wise Perturbation Profiles for Diagnosing Vision Robustness. *1st Workshop on Combining Theory and Benchmarks (CTB@ICML 2026)*. 2026.

[2] **Son Nguyen**, Xinyuan Liu, Ransalu Senanayake. CUPID in the Model Zoo: Online Matchmaking for Selecting Your Dream LLM. *International Conference on Machine Learning*. 2026.

[1] **Son Nguyen**, Xinyuan Liu, Ransalu Senanayake. LLM Routing as Reasoning: A MaxSAT View. *International Conference on Learning Representations Workshop on Logical Reasoning of Large Language Models*. 2026.

Research Experience

Research Assistant, MURGe-Lab UNC, Dr. Mohit Bansal — Aug 2026 – Present

- Improved RHAE on ARC-AGI-3 by 48× over parameter-matched StochasticGoose (ARC-AGI-3 Preview Competition winner) in a constrained budget setting.
- Completed 8× more games in ARC-AGI-3 by leveraging a factored transition model for MPC lookahead and maintaining persistent weights throughout the levels
- Implemented a neurosymbolic world model based on Dreamer-V3 and LLM-derived symbolic states that improves movement accuracy by 70%.

Undergraduate Researcher, LENS Lab ASU, Dr. Ransalu Senanayake — Aug 2025 – Aug 2026

- Built preference-based Bayesian optimization and multi-armed bandit pipelines to rank and match 125+ Large Language Models (LLMs) and image-generation models from pairwise user feedback in <0.1s; reduced cost by 37.61% and increased accuracy by 25% compared to the next best baseline.
- Formulated and characterized continuous attention as a finite-rank Gaussian Process (GP) that improves extrapolation Root Mean Square Error 43.0% over amortized GP and 7% over vanilla attention in 1s window.
- Fine-tuned LoRA adapters for Llama 3.1 8B Instruct on HelpSteer2 to synthesize extreme preference profiles across 5 reward dimensions; leveraged Retrieval-Augmented Generation (RAG) to augment models to domain experts across different subfields such as Finance, Healthcare, or Physics.

Researcher, MMLab University of Information Technology, Dr. Vinh-Tiep Nguyen — May 2026 – Aug 2026

- Developed an aesthetic-aware neurosymbolic framework for evaluating generated images, combining reinforcement learning and Bayesian optimization to discover interpretable symbolic rules over visual attributes.
- Formulated rule discovery as an MDP and trained a PPO agent to select visual predicates, rule templates, recalibration actions, and halt decisions, optimizing aesthetic prediction, consistency, novelty, and parsimony.
- Implemented a differentiable symbolic module with neural grounding, soft Boolean rules, learnable thresholds, and an MLP aggregator; generated semantic counterfactual explanations for visual model failures.

Academic Achievements

Bronze Medal – 2025 ICPC Rocky Mountain Regional Contest: second team awarded medal in ASU's history.

Dean's List – Fall 2025, Spring 2026

Third Prize – Vietnam National Olympiad in Artificial Intelligence (2025): rank 31

Third Prize – 5th Central–Central Highlands Informatics Olympiad (2024)

Second Prize – Ho Chi Minh City Informatics Olympiad (2022): Ranked 7
Third Prize – Tran Dai Nghia High School for the Gifted Research Competition (2022)

Teaching Experience

Undergraduate Learning Assistant, CSE 110: Principles of Programming, Arizona State University, Spring 2026

Other Activities

Technical Officer, The Software Developers Association (SoDA) at ASU — Jan 2026 – Aug 2026
Research Lead, Pet Food Bank Optimization for Lost Our Home Organization — Aug 2025 – Dec 2025

Standardized Test Scores

SAT: 1520 superscored
IELTS: 8.5 (Reading: 9; Listening: 9; Speaking: 7.5; Writing: 7.5)
Vietnam National University, Ho Chi Minh City 2025 Academic Competency Test (Session 1): 983/1200 (top 0.01%)
2022 Ho Chi Minh City High School Entrance Test: Composite Score 44 (top 0.001%), Perfect Score in Specialized Subject (6 students in total)